

Dicky Arnan

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Melbourne, VIC

Professional Summary

Master of Information Technology candidate at Monash University with a WAM of 89, specialising in cloud backend engineering, full-stack web development, and Android. Experienced in explaining technical concepts to peers and passionate about bridging coursework to real-world practice. Brings industry perspective from software engineering, telecommunications, and manufacturing leadership, with professional fluency in English, Indonesian, and Mandarin.

Education

Monash University – Master of Information Technology (WAM: 89, GPA: 3.9)	Jul 2024 – Present
Zhejiang University – Bachelor of Engineering in Mechatronics Engineering (GPA: 3.6)	Sep 2016 – Sep 2020

Project Experience

Relo'OZ: Suburb Recommendation Platform (AWS Lambda, API Gateway, PostgreSQL RDS, GitHub Actions, AWS SAM)
<https://relo-oz.app>

A web application that recommends Melbourne suburbs to vulnerable groups relocating to the city, based on personalised priorities across safety, housing affordability, transport, schools, and aged care data.

- Designed and deployed 8 AWS Lambda functions forming the complete REST API backend, each scoped to a single domain (crime, housing, transport, schools, aged care, suburb details, recommendations, CSV import) and declared in SAM template.yaml as Infrastructure as Code for reproducible deployment.
- Built a GitHub Actions CI/CD pipeline for automated deployment on push, and automated data ingestion via an S3-triggered Lambda that truncates and re-inserts datasets, eliminating upload risks and ensuring data consistency.
- Implemented multi-layer security including VPC private subnets, WAF rate limiting, and parameterised SQL queries across all endpoints to prevent SQL injection by design, with flood-test verification via Postman.

BirdEx: Wildlife Observation Android App (Kotlin, Android, Google Maps API, OpenWeather API, Room Database)
<https://github.com/nickchoi-monash/FIT5046-Birdex> (private)

A citizen science Android application for Australian birdwatchers to discover nearby hotspots, identify bird species through on-device AI sound recognition, and contribute to wildlife conservation data, aligned with SDG 15 (Life on Land).

- Implemented a sensor simulator that reads biometric and activity data from MIT-BIH Heart Rate and WISDM datasets via CSV, emitting through Kotlin Flow to trigger real-time alerts when user activity risks disturbing birds.
- Developed the Quest screen for location-based hotspot discovery with dropdown filtering and implemented bird species search functionality across the app's database.
- Built Google Maps API integration for in-app navigation to selected birdwatching locations with route guidance, and integrated OpenWeather API to display weather forecasts for planned birdwatching trips.

Work Experience

Software Engineer Intern, PT. Pro Sistimatika Automasi (PROSIA) – Indonesia Jan 2026 – Feb 2026
An Indonesian enterprise software and IT solutions provider with 20+ years of experience, delivering digital systems for government and enterprise clients and part of the Japan-based ITOCHU Techno-Solutions (CTC) Group.

- Developed internal full-stack modules for an Immigration Visa Processing System supporting 50+ offices for the Indonesian Directorate General of Immigration.
- Designed and implemented a backup server prototype with automated nightly sync, batching 500 records per call, with rollback support to ensure data integrity.
- Collaborated with a senior developer and system analyst to test, document, and demo solutions, ensuring smooth local operation prior to deployment.

Intern, Telstra – Melbourne Jun 2025 – Jul 2025
Telstra is Australia's largest telecommunications and technology company, providing mobile, internet, and enterprise services to millions of customers across Australia and internationally.

- Led research on Telstra's telecommunications strategy and market gaps, contributing to one of three innovative proposals adopted in the final project.
- Presented findings to 4 senior Telstra mentors across product, engineering, network, and data governance, receiving commendation for research depth, clarity, and practical recommendations.

Lead Manufacturing Engineer, PT. Bright Mobile Telecommunication – Indonesia May 2022 – June 2024
Indonesia's largest OPPO mobile phone manufacturing facility, with a production capacity of up to 2 million smartphones per month. OPPO is a global top 5 smartphone brand, dominating Southeast Asia with ~20% market share.

- Led a team of 8 in executing New Product Introduction (NPI) and New Machine Introduction (NMI) processes, ensuring on-time production with zero quality issues and downtime less than 10 minutes.
- Initiated ABB Robotics Automated Production Line, achieving a 15% cost reduction and 5% higher production rates.
- Implemented Lean Manufacturing strategies to boost factory efficiency by 10%.

Awards	
PT. Bright Mobile Telecommunication – The Best Employee 2023	Dec 2023
PT. Bright Mobile Telecommunication – The Best Team 2022	Dec 2022
Zhejiang University – Honorary Title of Outstanding International Student	2018 – 2019
Zhejiang University – International Student Scholarship	2016 – 2019
Organizational Experience	
Member , Hangzhou Indonesian Student Community (Culture and Sport Dept)	2015 – 2016
Vice-Chairman , High School Student Council	2013 – 2014
Vice-Chairman , High School Flag Raisers Force Batch 2013	2013 – 2015
Chairman , Junior High School Student Council	2010 – 2011
Skills	
Programming Languages: Python, Java, JavaScript, Kotlin, SQL	
Web and Frontend: Vue.js 3, HTML, CSS, RESTful APIs	
Cloud and Backend: AWS (Lambda, API Gateway, S3, DynamoDB, RDS, SAM), Node.js, Firebase, GitHub Actions	
Mobile: Android (Kotlin), TensorFlow Lite, Room Database, Retrofit	
Database: PostgreSQL, Oracle, MongoDB (basic)	
Tools: Git, Postman, VS Code	
Languages: Indonesian (Native), Mandarin (Fluent), English (Fluent)	